



Module-II Quantum Chemical Methods

Density Functional Theory, Geometry Optimization, Electronic Structure of Atoms and Molecules,

Ab Initio MD, Application,

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Quantum Mechanics

- **1998, Nobel Prize** in Chemistry → jointly to **Walter Kohn** for developing the foundations of **DFT** and **John Pople** for developing a quantum chemistry computer code for calculating the **electronic structure** of atoms and molecules
- **first** time that a Nobel prize in chemistry or physics was awarded for the development of a **numerical** method

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Recall



- **Wave-particle duality** is deeply embedded into the foundations of **quantum mechanics**.
- In the formalism of the theory, all the information about a particle is encoded in its **wave function**, a complex-valued function roughly analogous to the amplitude of a wave at each point in space.
- This function evolves according to **Schrödinger equation**.
- **A wave function (Ψ)** is a mathematical function that relates the location of an electron at a given point in space (identified by x, y, and z coordinates) to the amplitude of its wave, which corresponds to its energy.
- Thus each wave function is associated with a **particular energy E**.

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Recall



Schrödinger equation

$$\left[-\frac{\hbar^2}{2m} \sum_{i=1}^N \overset{(1)}{\nabla_i^2} + \sum_{i=1}^N \overset{(2)}{V(\mathbf{r}_i)} + \sum_{i=1}^N \sum_{j<i}^N \overset{(3)}{U(\mathbf{r}_i, \mathbf{r}_j)} \right] \psi = E\psi. \quad \hat{H}\Psi = E\Psi$$

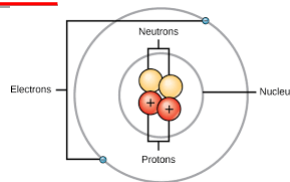
- In this equation, m is the electron mass
- The three terms in brackets → **1) Kinetic Energy of each electron**,
 - **2) Interaction energy between each electron & the collection of atomic nuclei**
 - **3) interaction energy between different electrons**
- For the Hamiltonian we have chosen Ψ is the electronic **wave function** → is **function** of each of the **spatial coordinates** of **each** of the **N electrons** → $\Psi = \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$
- **E** is the **ground state energy** of the electrons
- The ground-state energy is independent of time, so this is the **time-independent Schrödinger equation**

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DFT Calculations



- So, we can split our physical question into **two** pieces
- **1)** we solve for fixed positions of the **atomic nuclei**, the equations that describe the **electron motion**
- For a given set of electrons moving in the field of a set of nuclei, we find the **lowest energy configuration**, or **state**, of the electrons
- The lowest energy state is known as the **ground state** of the electrons
- **Born-Oppenheimer Approximation** : →
the separation of the nuclei and electrons into separate mathematical problems
- 2) If we have M nuclei positions ($R_1, \dots, R_M$), then we can express the ground-state energy, E , as a function of the positions of these nuclei, $E(R_1, \dots, R_M)$
- This function is known as the **adiabatic potential energy surface** of the atoms



- 1) One describes the **motion of the nuclei**.
- 2) The other one describes the **motion of the electrons** around the nuclei whose positions are fixed.

Wave-Function-Based Methods



- A second fundamental classification of quantum chemistry calculations can be made according to the **quantity** that is being **calculated**.
- *DFT the aim is to compute the **electron density**, not the **electron wave function***
- There are many methods, however, where the object of the calculation is to compute the full electron wave function.
- wave-function-based methods → given infinite computer time, can converge to the exact solution of the Schrödinger equation

Hartree-Fock

- Independent electron approximation

Density Functional Theory

- Consider electron density instead of the wavefunction

Quantum Monte Carlo

- Statistical method



Hurdles of solving the Schrödinger equation for practical materials

1. the **electronic wave function** Ψ is → is function of each of the spatial coordinates of each of the **N electrons** → $\Psi = \Psi(r_1, \dots, r_N)$

Hartree Approximation



- Neglecting the correlation – assumes the electrons are uncorrelated
- Total wave function is written as the product of single electron wave functions

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \psi_1(\mathbf{r}_1) \psi_2(\mathbf{r}_2) \dots \psi_N(\mathbf{r}_N)$$

Hartree product

approximating Ψ as a product of individual electron wave functions, $\Psi = \psi_1(\mathbf{r}) \psi_2(\mathbf{r}), \dots, \psi_N(\mathbf{r})$

- Variational principle leads to Hartree equation

$$\left[-\frac{1}{2} \nabla_i^2 + V_{\text{ion}} + V_H(\mathbf{r}_i)\right] \psi_i(\mathbf{r}_i) = \epsilon_i \psi_i(\mathbf{r}_i)$$

$$V_H(\mathbf{r}_i) = 2 \sum_{j \neq i} \int \frac{|\psi_j(\mathbf{r}_j)|^2}{|\mathbf{r}_i - \mathbf{r}_j|} d\mathbf{r}_j$$

Hartree potential,
average Coulomb potential
from other electrons

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Recall

DFT Calculations

Hurdles of solving the Schrödinger equation for practical materials



- **Hamiltonian** defining **electron –electron** interactions (a **many-body problem**)
- Measurable quantity $\rightarrow$ is the **probability** that the N electrons are at a particular set of coordinates, $\mathbf{r}_1, \dots, \mathbf{r}_N$
- the **density** of electrons at a particular position in space, $n(\mathbf{r})$ in terms of the individual electron wave functions
- Functional described by the Hohenberg – Kohn theorem is in terms of the single-electron wave functions, $\Psi(\mathbf{r})$.
- The **energy functional** can be written as

$$\text{Probability } P = \Psi^*(\mathbf{r}_1, \dots, \mathbf{r}_N) \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$$

$$n(\mathbf{r}) = 2 \sum_i \psi_i^*(\mathbf{r}) \psi_i(\mathbf{r})$$

$$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$$

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Recall

DFT Calculations

Hurdles of solving the Schrödinger equation for practical materials



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$$n(\mathbf{r}) = 2 \sum_i \psi_i^*(\mathbf{r}) \psi_i(\mathbf{r})$$

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- **Theorem I: density as a basic variable**
- The total energy of a many-body system is a unique functional of electron density
- The total energy functional can be written as

$$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$$

$\uparrow$ interaction with nuclei and any other field

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- Theorem I: density as a basic variable
- The total energy of a many-body system is a unique functional of electron density
- The total energy functional can be written as

$$E[n] = F[n] + \int v_{\text{ext}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}$$

interaction with nuclei
and any other field

- Universal functional $F[n]$ is independent of external potential – but **unknown!**

$$F[n] = T[n] + U[n]$$

Hohenberg-Kohn theorem

- Theorem II: variational for the energy in term of density
- The functional $E[n(\mathbf{r})]$ has its minimum when electron density is at the equilibrium electron density $n_0(\mathbf{r})$
- The minimum of the energy E_0 is exactly equivalent to the true ground-state energy

$$E_0 \equiv \min E[n(\mathbf{r})] \equiv E[n_0(\mathbf{r})]$$

Hohenberg and Kohn Theorem

DFT: From Wave Functions To Electron Density

- The “known” terms include four contributions:

$$E_{\text{known}}[\{\psi_i\}] = -\frac{\hbar^2}{2m} \sum_i \int \psi_i^* \nabla^2 \psi_i d^3r + \int V(\mathbf{r})n(\mathbf{r}) d^3r$$
- 1) the electron kinetic energies,
- 2) the Coulomb interactions between the electrons and the nuclei,

$$+ \frac{e^2}{2} \int \int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' + E_{\text{ion}}.$$
- 3) the Coulomb interactions between pairs of electrons, and
- 4) the Coulomb interactions between pairs of nuclei.
- The other term in the complete energy functional, $E_{\text{xc}}[\{\psi_i\}]$, is the **exchange - correlation functional**,
- The Kohn -Sham equations $\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r})$

Kohn -Sham equations

DFT

in the following algorithm:

1. Define an initial, trial electron density, $n(\mathbf{r})$
2. Solve the Kohn -Sham equations defined using the trial electron density to find the single-particle wave functions, $\Psi(\mathbf{r})$.
3. Calculate the electron density defined by the Kohn - Sham single-particle wave functions from step 2, $n_{\text{KS}}(\mathbf{r}) = 2 \sum_i \psi_i^*(\mathbf{r})\psi_i(\mathbf{r})$
4. Compare the calculated electron density, $n_{\text{KS}}(\mathbf{r})$, with the electron density used in solving the Kohn -Sham equations, $n(\mathbf{r})$.
- If the two densities are the same, then this is the ground-state electron density, and it can be used to compute the total energy.
- If the two densities are different, then the trial electron density must be updated in some way.
- Once this is done, the process begins again from step 2.

EXCHANGE – CORRELATION FUNCTIONAL



- finding the ground-state energy of the Schrödinger equation,
- but this is extremely **difficult** because this is a **many-body** problem.
- The beautiful results of **Kohn**, **Hohenberg**, and **Sham** showed us that the ground state we seek can be found by **minimizing the energy** of an **energy functional**, and
- that this can be achieved by finding a **self-consistent solution** to a set of **single-particle** equations.
- There is just **one** critical **complication** in this otherwise beautiful formulation: to solve the Kohn – Sham equations
- we must specify the **exchange – correlation** function, $E_{xc}[\{\Psi_i\}]$.

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EXCHANGE – CORRELATION FUNCTIONAL



- Exchange
 - Keep electrons of same spin away from each other (Pauli exclusion)
- Correlation
 - Keep electrons away from each other due to Coulombic forces
- Finding E_{xc} functional is the great challenge, but approximate functionals work:
 - LDA
 - GGA
 - Hybrid functional
 - EXX

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EXCHANGE – CORRELATION FUNCTIONAL



- the true form of the exchange – correlation functional whose existence is guaranteed by the Hohenberg –Kohn theorem is simply not known.
- Exchange and/or correlation functionals are often written in the following form:

$$E_{xc}[n] = \int n(\mathbf{r}) \epsilon_{xc}[n(\mathbf{r})] d\mathbf{r}$$
 - Where the exchange-correlation energy density

$$\epsilon_{xc}[n(\mathbf{r})]$$
 represents the **energy per electron**
 - The density

$$n(\mathbf{r})$$
 as usual, represents the number of **electrons per unit volume**

$$V_{xc}(\mathbf{r}) = \frac{\delta E_{xc}(\mathbf{r})}{\delta n(\mathbf{r})}$$

a “functional derivative” of the exchange – correlation energy

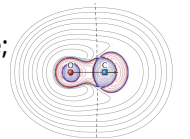
EXCHANGE – CORRELATION FUNCTIONAL



- the true form of the exchange – correlation functional whose existence is guaranteed by the Hohenberg –Kohn theorem is simply not known.
- Fortunately, there is one case where this functional can be derived exactly: **the uniform electron gas**.
- In this situation, the electron density is constant at all points in space; that is, **$n(\mathbf{r}) = \text{constant}$** .
- This situation may appear to be of **limited** value in any real material since it is variations in electron density that **define chemical bonds** and generally make materials interesting
- we set the **exchange – correlation potential** at each position to be the known exchange –correlation potential from the uniform electron gas at the electron density observed at that position

$$V_{xc}(\mathbf{r}) = \frac{\delta E_{xc}(\mathbf{r})}{\delta n(\mathbf{r})}$$

a “functional derivative” of the exchange – correlation energy

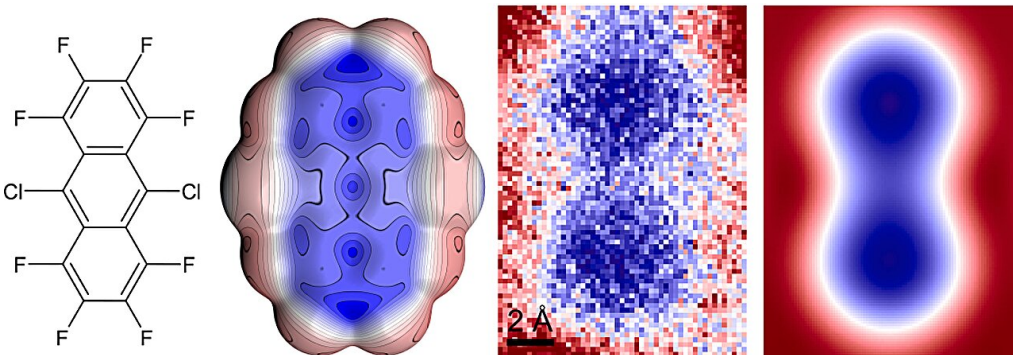


electron density

$$V_{xc}(\mathbf{r}) = V_{xc}^{\text{electron gas}}[n(\mathbf{r})]$$

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<https://www.nature.com/articles/s41467-023-40593-3>

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EXCHANGE – CORRELATION FUNCTIONAL



$$V_{XC}(\mathbf{r}) = V_{XC}^{\text{electron gas}}[n(\mathbf{r})]$$

- This approximation uses only the local density to define the approximate exchange – correlation functional, so it is called the local density approximation (LDA).
- The LDA gives us a way to completely define the Kohn–Sham equations,
- but the results from these equations do not exactly solve the true Schrödinger equation because we are not using the true exchange – correlation functional
- The best known class of functional after the LDA uses information about the local density $E_{xc}^{\text{LDA}}[n] = E_x^{\text{LDA}}[n] + E_c^{\text{LDA}}[n]$
- Assume that the exchange-correlation energy density at every position in space for the molecule is the same as it would be for the uniform electron gas (UEG) having the same density as found at that position
- The expression for the exchange energy is simple $E_x^{\text{LDA}}[n] = C \int n^{\frac{4}{3}}(\mathbf{r}) d\mathbf{r}$

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local density approximation (LDA)

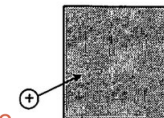
- The simplest but surprisingly good approximation
- Assume that the charge density varies slowly; **each small volume looks like a uniform electron gas**
- $E_{xc}[n]$ can be obtained by integrating uniform e⁻ gas results

$$E_{xc}^{\text{LDA}}[n] = \int n(\mathbf{r}) \epsilon_{xc}(n) d\mathbf{r}$$

- $\epsilon_{xc}(n)$ is the exchange-correlation energy per electron in homogenous electron gas at density n , which can be calculated

- More generally, formulation for spin density

$$E_{xc}^{\text{LSDA}}[n_{\uparrow}, n_{\downarrow}] = \int n(\mathbf{r}) \epsilon_{xc}(n_{\uparrow}, n_{\downarrow}) d\mathbf{r}$$



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EXCHANGE – CORRELATION FUNCTIONAL



- The best known class of functional after the LDA uses information about the local electron density and the **local gradient in the electron density**; $V_{XC}(\mathbf{r}) = V_{XC}^{\text{electron gas}}[n(\mathbf{r})]$
- this approach defines a **generalized gradient approximation (GGA)**.
- An improvement in the accuracy provided by the LDA can be obtained by Generalised Gradient Approximation (GGA) functionals
- These depend not just on the value of the density at a point (as in the LDA case) but also on its gradient. So we have $E_{xc}^{\text{GGA}}[n] = \int n(\mathbf{r}) \epsilon_{xc}(n(\mathbf{r}), |\nabla n(\mathbf{r})|) d\mathbf{r}$
- Most GGA functionals are constructed in the form of a correction term which is added to the LDA functional $\epsilon_{xc/c}^{\text{GGA}}[n] = \epsilon_{xc/c}^{\text{LDA}}[n] + \Delta \epsilon_{xc/c} \left[\frac{|\nabla n(\mathbf{r})|}{n^{\frac{4}{3}}(\mathbf{r})} \right]$

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- Density gradient expansion
- Take the value of the density at each point as well as the magnitude of the gradient of the density

$$E_{xc}^{GGA}[n] = \int n(\mathbf{r}) \varepsilon_{xc}(n, \nabla n) d\mathbf{r}$$

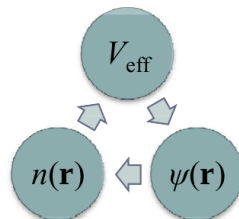
- Make improvement over LDA for many cases
 - E.g. improve predicted binding and dissociation energies

$$V_{xc}(\mathbf{r}) = V_{xc}^{\text{electron gas}}[n(\mathbf{r})]$$

- The best known class of functional after the LDA uses information about the local electron density and the **local gradient in the electron density**;
- this approach defines a **generalized gradient approximation** (GGA).
- there are many ways in which information from the gradient of the electron density can be included in a GGA functional,
- there are a large number of distinct GGA functionals.
- Two of the most widely used functionals in calculations involving solids are the Perdew –Wang functional (PW91) and the Perdew –Burke– Ernzerhof functional (PBE).
- It is tempting to think that because the GGA includes more physical information than the LDA it must be more accurate.
- Unfortunately, this is not always correct.

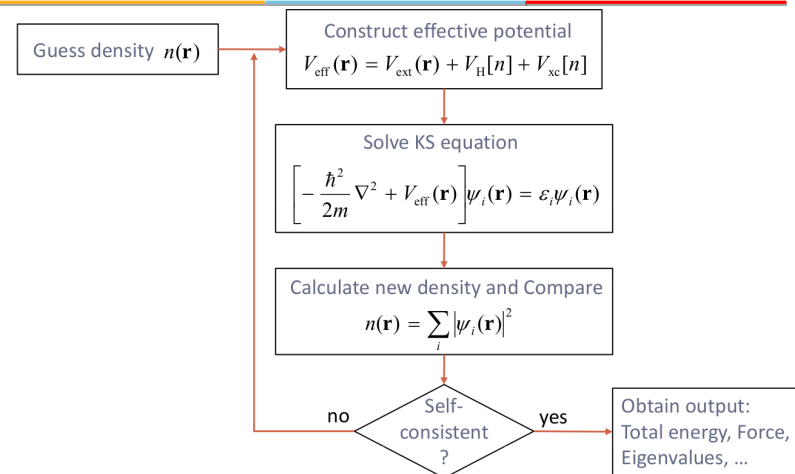
$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{xc}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r})$$

- Because the effective potential V_{eff} depends upon $n(\mathbf{r})$, and hence $\psi(\mathbf{r})$, which is the solution of the equation



- The Kohn-Sham equations must be solved by an iterative technique; **self-consistent field method**

Self-consistent calculation procedure

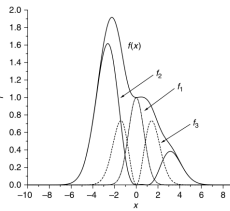


Localized and Spatially Extended Functions



- One useful way to classify quantum chemistry calculations is according to the types of functions they use to represent their solutions.
- Broadly, these methods use either spatially localized functions or spatially extended functions.

- an example of a spatially localized function where $f_1(x) = \exp(-x^2)$,
 $f_2(x) = x^2 \exp(-x^2/2)$,
 $f_3(x) = \frac{1}{10} x^2 (1 - x^2) \exp(-x^2/4)$,
 $f(x) = f_1(x) + f_2(x) + f_3(x)$.
- All of these functions rapidly approach zero for large values of $|x|$
- Functions like this are entirely appropriate for representing the wave function or electron density of an isolated atom
- could include more information in this final function by including more individual functions
- could build up functions that describe multiple atoms simply by using an appropriate set of localized functions for each individual atom



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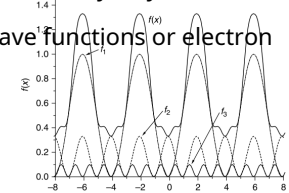
Localized and Spatially Extended Functions



- Spatially localized functions** are an extremely useful framework for thinking about the quantum chemistry of **isolated molecules** because the **wave functions** of **isolated molecules** really do decay to **zero far away from the molecule**.
- But what if we are interested in a bulk material such as the atoms in solid silicon or the atoms beneath the surface of a metal catalyst?
- We could still use **spatially localized functions** to describe **each atom** and **add up** these functions to describe the **overall** material, but this is certainly **not** the only way forward.
- A useful alternative is to use periodic functions to describe the wave functions or electron densities.
- The resulting function is periodic
- bulk materials: electron density and wave function really are spatially periodic functions**

$$f(x) = f_1(x) + f_2(x) + f_3(x),$$

where $f_1(x) = \sin^2\left(\frac{\pi x}{4}\right)$,
 $f_2(x) = \frac{1}{3} \cos^2\left(\frac{\pi x}{2}\right)$,
 $f_3(x) = \frac{1}{10} \sin^2(\pi x)$.



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Wave-Function-Based Methods:

what, exactly, is a wave function?



- We must first describe the common ways in which the wave function is described
- wave function of an **N-particle system** is an **N-dimensional** function
- We want our **wave functions** to provide a quantum mechanical description of a system of **N electrons**,
- these wave functions must satisfy several mathematical properties exhibited by **real electrons**.
- Pauli exclusion principle prohibits two electrons with the same spin from existing at the same physical location simultaneously
- These properties to also exist in any approximate form of the wave function that we construct

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Wave-Function-Based Methods:

Hartree –Fock Method



- Electrons are fermions; obey the Pauli exclusion principle and Fermi statistics
 - Wave function changes sign when the coordinates of two electrons are interchanged
 - Wave function is zero if two electrons are in the same state
 - Wave function is written to be an **antisymmetrized product** (Slater Determinant)

$$\Psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{r}_1) & \psi_1(\mathbf{r}_2) & \cdots & \psi_1(\mathbf{r}_N) \\ \psi_2(\mathbf{r}_1) & \psi_2(\mathbf{r}_2) & \cdots & \psi_2(\mathbf{r}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_N(\mathbf{r}_1) & \psi_N(\mathbf{r}_2) & \cdots & \psi_N(\mathbf{r}_N) \end{vmatrix}$$

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Wave-Function-Based Methods:

Hartree –Fock Method



- we would like to approximate the wave function of N electrons.
- Let us assume for the moment that the electrons have no effect on each other (neglect electron–electron interactions).
- If this is true, the Hamiltonian for the electrons may be written as $H = \sum_{i=1}^N h_i$
- h_i describes the kinetic and potential energy of electron i.
- Schrödinger equation for just one electron based on this Hamiltonian,
- the solutions would satisfy $h\chi = E\chi$
- The eigenfunctions defined by this equation are called spin orbitals
- For each single-electron equation there are multiple eigenfunctions

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Wave-Function-Based Methods:

Hartree –Fock Method

$$H = \sum_{i=1}^N h_i \quad h\chi = E\chi$$



- this defines a set of **spin orbitals** $\chi_j(\mathbf{x}_i)$ ($j = 1, 2, \dots$) where $\mathbf{x}_i$ is a vector of **coordinates** that defines the **position** of **electron i** and its **spin state** (up or down)
- the energy of spin orbital $\chi_j(\mathbf{x}_i)$ by E_j
- It is useful to label the spin orbitals so that the orbital with **j = 1** has the **lowest** energy, the orbital with **j = 2** has the **next** highest energy, and so on.
- When the total Hamiltonian is simply a sum of one-electron operators, h_i , it follows that the eigenfunctions of H are products of the one-electron spin orbitals:
$$\psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \chi_{j_1}(\mathbf{x}_1)\chi_{j_2}(\mathbf{x}_2) \cdots \chi_{j_N}(\mathbf{x}_N)$$
- The energy of this wave function is the sum of the spin orbital energies,
 $E = E_{j_1} + \dots + E_{j_N}$
- this approximation to the N-electron wave function, the **Hartree** product

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Wave-Function-Based Methods:

Hartree –Fock Method



$$\psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \chi_{j_1}(\mathbf{x}_1)\chi_{j_2}(\mathbf{x}_2) \cdots \chi_{j_N}(\mathbf{x}_N)$$

- Unfortunately, the **Hartree** product does not satisfy all the important criteria for wave functions.
- Because electrons are **fermions**, the wave function must change **sign** if two electrons change places with each other.
- This is known as the **antisymmetry principle**.
- Exchanging two electrons does not change the sign of the Hartree product, which is a serious drawback.
- We can obtain a better approximation to the wave function by using a **Slater determinant**.
- In a Slater determinant, the N-electron wave function is formed by combining one-electron wave functions in a way that satisfies the antisymmetry principle.

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Wave-Function-Based Methods:

Hartree –Fock Method:Slater determinant



- In a **Slater determinant**, the N-electron wave function is formed by combining one-electron wave functions in a way that satisfies the **antisymmetry principle**.
- This is done by expressing the overall wave function as the determinant of a **matrix of single-electron wave functions**.
- It is best to see how this works for the case of **two electrons**.
- For two electrons, the Slater determinant is $\rightarrow \rightarrow \psi(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}} \det \begin{bmatrix} \chi_j(\mathbf{x}_1) & \chi_j(\mathbf{x}_2) \\ \chi_k(\mathbf{x}_1) & \chi_k(\mathbf{x}_1) \end{bmatrix}$
$$= \frac{1}{\sqrt{2}} [\chi_j(\mathbf{x}_1)\chi_k(\mathbf{x}_2) - \chi_j(\mathbf{x}_2)\chi_k(\mathbf{x}_1)]$$
- coefficient of $(1/\sqrt{2})$ is simply a normalization factor
- physical description of electron exchange implicitly; it changes sign if two electrons are exchanged

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Wave-Function-Based Methods:

Hartree –Fock Method:Slater determinant



- This expression has other advantages.
- For example, it does not distinguish between electrons
- it disappears if two electrons have the same coordinates or
- if two of the one-electron wave functions are the same.
- This means that the Slater determinant satisfies the conditions of the Pauli exclusion principle.
- The Slater determinant may be generalized to a system of N electrons easily;
- it is the determinant of an N N matrix of single-electron spin orbitals.
- By using a Slater determinant, we are ensuring that our method for solving the Schrödinger equation will include exchange.

$$\begin{aligned}\psi(\mathbf{x}_1, \mathbf{x}_2) &= \frac{1}{\sqrt{2}} \det \begin{bmatrix} \chi_j(\mathbf{x}_1) & \chi_j(\mathbf{x}_2) \\ \chi_k(\mathbf{x}_1) & \chi_k(\mathbf{x}_1) \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} [\chi_j(\mathbf{x}_1)\chi_k(\mathbf{x}_1) - \chi_j(\mathbf{x}_2)\chi_k(\mathbf{x}_1)]\end{aligned}$$

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Wave-Function-Based Methods:

Hartree –Fock Method:Slater determinant



This is done by expressing the overall wave function as the determinant of a *matrix of single-electron wave functions*.

For two electrons, the Slater determinant is

$$\begin{aligned}\psi(\mathbf{x}_1, \mathbf{x}_2) &= \frac{1}{\sqrt{2}} \det \begin{bmatrix} \chi_j(\mathbf{x}_1) & \chi_j(\mathbf{x}_2) \\ \chi_k(\mathbf{x}_1) & \chi_k(\mathbf{x}_1) \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} [\chi_j(\mathbf{x}_1)\chi_k(\mathbf{x}_1) - \chi_j(\mathbf{x}_2)\chi_k(\mathbf{x}_1)]\end{aligned}$$

The coefficient of (1/√2) is simply a normalization factor.

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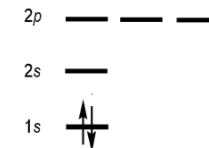
Wave-Function-Based Methods:

Hartree –Fock Method:Slater determinant



- In a *Slater determinant*, the N-electron wave function is formed by combining one-electron wave functions in a way that satisfies the *antisymmetry principle*.

- **Slater determinant satisfies the conditions of the Pauli exclusion principle:** it disappears if two electrons have the same coordinates or if two of the one-electron wave functions are the same
- it is the determinant of an N x N matrix of single-electron spin orbitals



- physical description of electron exchange implicitly; it changes sign if two electrons are exchanged

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Wave-Function-Based Methods:

Hartree –Fock Method



- we would like to approximate the wave function of N electrons.
- Let us assume for the moment that the electrons have no effect on each other (neglect electron-electron interactions).
- If this is true, the Hamiltonian for the electrons may be written as $H = \sum_{i=1}^N h_i$
- h_i describes the kinetic and potential energy of electron i.
- Schrödinger equation for just one electron based on this Hamiltonian,
- the solutions would satisfy $h\chi = E\chi$
- The eigenfunctions defined by this equation are called spin orbitals
- For each single-electron equation there are multiple eigenfunctions

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Wave-Function-Based Methods:

Hartree –Fock Method

$$h\chi = E\chi$$



- Using a **Slater determinant**, we are ensuring that our method for solving the **Schrödinger equation** will include **exchange**
- electron correlation**: Any interesting system the electrons interact with one another
- how these interactions are approximated?
- In HF calculation, we fix the positions of the atomic nuclei and aim to determine the wave function of N-interacting electrons.

The Schrödinger equation for each electron is written as

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) \right] \chi_j(\mathbf{x}) = E_j \chi_j(\mathbf{x})$$

$$V_H(\mathbf{r}) = e^2 \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 r'$$

Difference with K-S Equation: is the additional **exchange – correlation** potential that appears in the Kohn – Sham equations

single electron “feels” the effect of other electrons only as an average, rather than feeling the instantaneous repulsive forces generated as electrons become close in space

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Wave-Function-Based Methods:

Hartree –Fock Method

To complete our description of the HF method, we have to define how the solutions of the single-electron equations above are expressed and how these solutions are combined to give the **N-electron wave function**.

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) \right] \chi_j(\mathbf{x}) = E_j \chi_j(\mathbf{x})$$

The HF approach assumes that the complete wave function can be approximated using a single Slater determinant.

$$\Psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{r}_1) & \psi_1(\mathbf{r}_2) & \cdots & \psi_1(\mathbf{r}_N) \\ \psi_2(\mathbf{r}_1) & \psi_2(\mathbf{r}_2) & \cdots & \psi_2(\mathbf{r}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_N(\mathbf{r}_1) & \psi_N(\mathbf{r}_2) & \cdots & \psi_N(\mathbf{r}_N) \end{vmatrix}$$

Means, N lowest energy spin orbitals of the single-electron equation are found

For N=2

$$\Psi(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}} \{ \chi_1(\mathbf{x}_1) \chi_2(\mathbf{x}_2) - \chi_1(\mathbf{x}_2) \chi_2(\mathbf{x}_1) \}$$

$$= \frac{1}{\sqrt{2}} \begin{vmatrix} \chi_1(\mathbf{x}_1) & \chi_2(\mathbf{x}_1) \\ \chi_1(\mathbf{x}_2) & \chi_2(\mathbf{x}_2) \end{vmatrix}$$

the total wave function is formed from the Slater determinant of these spin orbitals

$\chi_j(\mathbf{x})$ for $j = 1, \dots, N$,

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Wave-Function-Based Methods:

Hartree –Fock Method

$$H = \sum_{i=1}^N h_i$$

$$h\chi = E\chi$$



- Minimize the energy with respect to Slater determinant leads to the **Hartree-Fock equations**

$$\left[-\frac{1}{2} \nabla_i^2 + V_{\text{ion}} + V_H(\mathbf{r}_i) + V_X(\mathbf{r}_i) \right] \psi_i(\mathbf{r}_i) = \varepsilon_i \psi_i(\mathbf{r}_i)$$

$$V_X(\mathbf{r}_i) = -\frac{1}{2} \left[\sum_j \int \frac{\psi_j^*(\mathbf{r}_j) \psi_i(\mathbf{r}_j)}{|\mathbf{r}_i - \mathbf{r}_j|} d\mathbf{r}_j \right] \frac{\psi_j(\mathbf{r}_i)}{\psi_i(\mathbf{r}_i)}$$

- Adding Fermionic nature of electrons to the Hartree method give rise to an effective potential called **exchange interaction**
- still neglect the correlation effect, so large deviation from the experiment results

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- this defines a set of **spin orbitals** $\chi_j(\mathbf{x}_i)$ ($j = 1, 2, \dots$) where $\mathbf{x}_i$ is a vector of **coordinates** that defines the **position** of **electron i** and its **spin state** (up or down)
 - the energy of spin orbital $\chi_j(\mathbf{x}_i)$ by E_j
 - It is useful to label the spin orbitals so that the orbital with **j = 1** has the **lowest** energy, the orbital with **j = 2** has the **next** highest energy, and so on.
 - When the total Hamiltonian is simply a sum of one-electron operators, **h_i** , it follows that the eigenfunctions of H are products of the one-electron spin orbitals:
- $$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \chi_{j_1}(\mathbf{x}_1) \chi_{j_2}(\mathbf{x}_2) \cdots \chi_{j_N}(\mathbf{x}_N)$$
- The energy of this wave function is the sum of the spin orbital energies, $E = E_{j_1} + \dots + E_{j_N}$
 - this approximation to the N-electron wave function, the **Hartree product**

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